ENCODE LONDON HACKATHON · 2026

DRY RUN

Practise your real driving test, on your real test roads.

dry-runserver-production.up.railway.app

THE PROBLEM

Learners walk into the test blind

< 50%

of UK learner drivers pass the practical test

Months

of waiting for another slot after a fail

New roads

most learners take their test on roads they’ve never driven

WHAT IT DOES

Your test centre, in your browser

Your test area

Type a test centre or postcode. It builds the real streets and test-style routes through real roundabouts, lights and give-ways.

Mock test

A spoken examiner gives directions and a ‘tell me’ question. Faults are marked like the DVSA sheet; pass or fail at the end.

Debrief

Every fault explained, citing the Highway Code — while you drive and in full at the end.

Practice mode

Spoken tips after each fault, repeat a junction, and ‘your weak junctions’ from past drives.

HOW A DRIVE WORKS

From postcode to debrief

1

Pick a centre

Real streets from OpenStreetMap + Overture buildings

2

Drive in 3D

Three.js + Rapier physics, in the browser

3

Examiner speaks

Directions and a ‘tell me’ question, by voice

4

Rules mark you

11 rules: driving, serious, dangerous

5

AI explains

Each fault, with Highway Code citations

WHY YOU CAN TRUST IT

The AI can’t mark you, and can’t invent a rule.

Code marks faults

A rule engine decides every fault. Only exception: the ‘tell me’ answer, judged against the official DVSA answer.

Citations are checked

An explanation shows only if it cites real Highway Code rules it was given. Otherwise it’s sent back once, then dropped.

No personal data

No accounts, no camera — a random browser ID. If AI or network fails, you keep driving and nothing is marked.

UNDER THE HOOD · MONGODB ATLAS

One data platform for the whole drive

RAG: FINDING THE RULE FOR EACH FAULT



Live saving: every session and fault stored as it happens, geo-indexed

Aggregation pipeline: finds your weak junctions

Stream Processing: live fault counts per junction

STATS19: DfT road-collision data for the same junctions

UNDER THE HOOD · GEMINI

Three models, three jobs

The ‘tell me’ question

gemini-3.8-live (Live API)

Spoken question and answer, with a one-use token so the API key stays on the server.

The debrief

gemini-4-argon

Writes fault explanations and the end-of-test summary, checked by the citation checker.

The examiner’s voice

gemini-3.8-flash-tts

52 fixed sentences made ahead, so driving never waits for the network. Plus the spoken summary.

TRY IT NOW

TAKE A DRY RUN

dry-runserver-production.up.railway.app

Chrome on a laptop, sound on. Pre-built centres: Wanstead, Goodmayes and Chingford.

github.com/dastardlycole/dry-run